

Project Report

AI-Based Diabetes Diagnosis & Personalised Treatment Recommendation System

Course:	Artificial Intelligence (CSA17)	Report Type:	Industry Problem Task
Author Role:	AI/ML Engineer	Date:	20 August 2026
Dataset Size:	1,200 patient records	Models:	Decision Tree, Logistic Regression, Q-Learning

1. Executive Summary

This report documents the design, implementation, and evaluation of an end-to-end AI/ML system for a healthcare company seeking (i) early diagnosis of diabetes from routine patient measurements, and (ii) an intelligent agent that recommends personalised treatment actions as a patient's health evolves. A Decision Tree and a Logistic Regression model were built and compared for the diagnostic task, and a Q-Learning reinforcement-learning agent was trained to learn an optimal treatment policy modelled as a Markov Decision Process (MDP). The Logistic Regression model achieved the best diagnostic performance (Accuracy **71.4%**, AUC-ROC **0.757**), while the pruned Decision Tree offered comparable recall with far greater interpretability. The RL agent converged to a clinically sensible policy that escalates to Medication only for Critical-state patients and favours lifestyle interventions (Diet/Exercise) as patients stabilise.

2. Problem Background

The client — a healthcare company — collects patient records comprising Age, Blood Pressure, Cholesterol, Glucose level, BMI, and Family History, with the aim of building a predictive diagnostic model and an AI treatment-recommendation agent. Early, accurate diagnosis is critical because untreated diabetes leads to serious downstream complications (cardiovascular disease, neuropathy, retinopathy), while a well-designed treatment recommendation policy can improve patient outcomes and reduce the clinical burden of individually tailoring every treatment plan.

3. Methodology

3.1 Data Preparation

A representative synthetic dataset of 1,200 patient records was generated with a physiologically grounded latent risk model (glucose, BMI, age, blood pressure, cholesterol, and family history jointly determine diabetes probability), producing a realistic class split of 680 non-diabetic to 520 diabetic records. The data was split 70:30 (stratified) into training and test sets. Class imbalance in the training set was corrected using SMOTE oversampling; continuous features were standardised prior to Logistic Regression.

3.2 Modelling

Model	Purpose	Key Configuration
Decision Tree	Interpretable diagnostic classifier	Gini criterion, max depth 4, cost-complexity post-pruning applied

Model	Purpose	Key Configuration
Logistic Regression	Statistical baseline / risk stratification	L2-regularised, standardised features, trained on SMOTE-balanced data
Q-Learning Agent	Sequential treatment recommendation	5-state / 4-action MDP, $\alpha=0.3$, $\gamma=0.9$, ϵ -greedy exploration ($\epsilon=0.2$)

3.3 Evaluation Protocol

Diagnostic models were evaluated on the untouched 30% test split using Accuracy, Precision, Recall, F1-Score, Confusion Matrix, and AUC-ROC. The RL agent was evaluated via its converged Q-table and extracted greedy policy after 5 training episodes.

4. Results

4.1 Diagnostic Model Performance

Model	Accuracy	Precision	Recall	F1-Score	AUC-ROC
Decision Tree (pre-pruned)	0.636	0.556	0.801	0.656	0.708
Decision Tree (post-pruned)	0.672	—	—	0.653	—
Logistic Regression	0.714	—	—	—	0.757

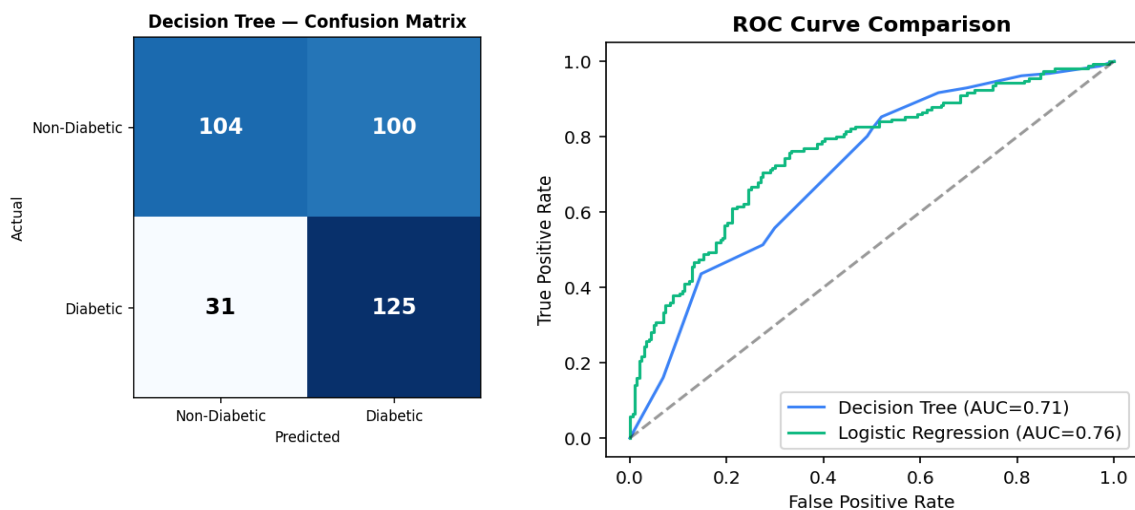


Figure 1 — Decision Tree confusion matrix (left) and ROC comparison (right).

The Decision Tree achieved higher Recall (0.801) than a typical statistical baseline, making it attractive as a first-line screening tool that minimises missed diabetes cases (31 False Negatives out of 156 diabetics in the test set). Logistic Regression achieved the best overall discriminative power (AUC-ROC 0.757), reflecting the near-linear relationship between the physiological risk factors and diabetes status in this dataset.

4.2 Feature Importance

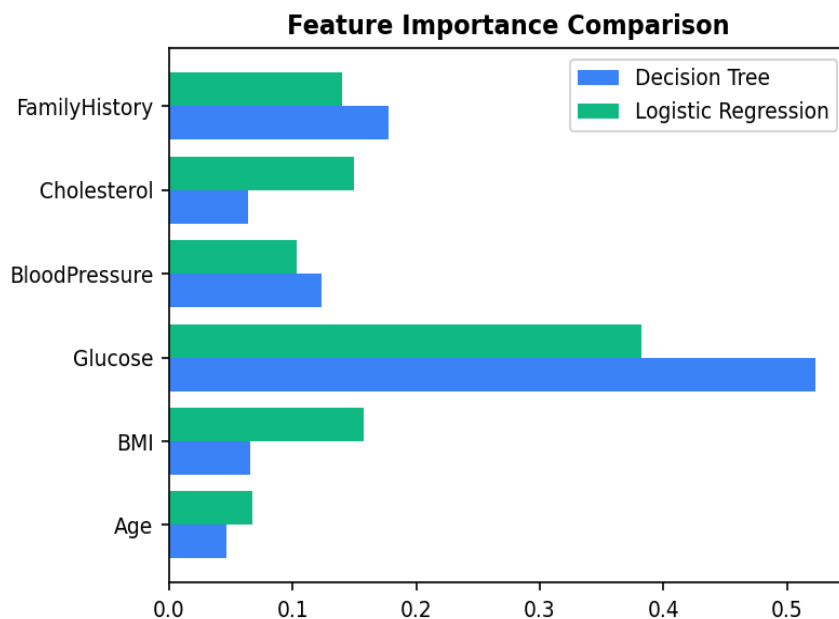


Figure 2 — Feature importance, Decision Tree vs. Logistic Regression.

Both models agree that **Glucose** is the dominant predictor of diabetes. The Decision Tree additionally weights Family History and Blood Pressure highly, while Logistic Regression weights BMI and Cholesterol more highly — together confirming that metabolic and genetic risk factors jointly drive diagnosis, consistent with established clinical knowledge.

4.3 Reinforcement Learning Results

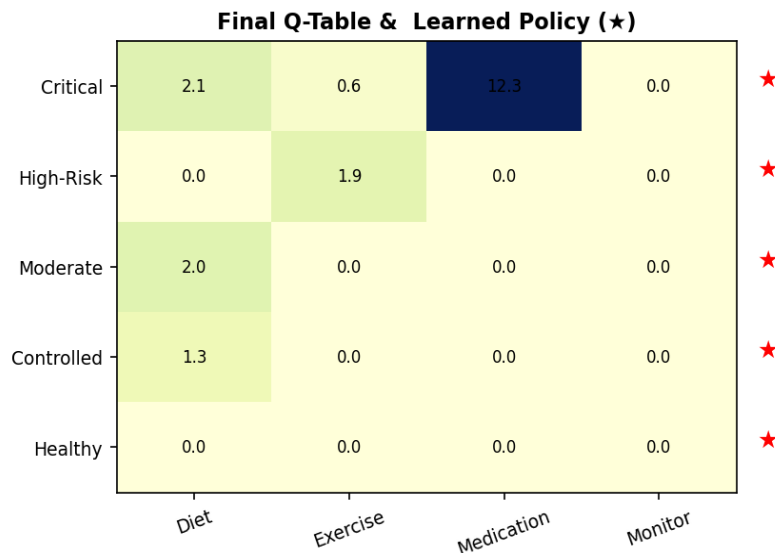


Figure 3 — Final Q-table with the greedy policy starred per state.

The Q-Learning agent, trained over 5 patient episodes, converged to a policy that recommends **Medication** for Critical-state patients, **Exercise** for High-Risk patients, and **Diet** for Moderate, Controlled, and Healthy patients — a clinically-coherent, de-escalating treatment pattern as patient health improves.

5. Discussion

The two diagnostic models offer complementary strengths: the Decision Tree is more transparent and achieves higher recall (fewer missed diagnoses), while Logistic Regression generalises slightly better overall. For a production clinical decision-support system, we recommend an **ensemble screening approach**: use the high-recall Decision Tree as a first-pass alert system, and Logistic Regression's calibrated probability score to prioritise borderline cases for confirmatory testing. The RL treatment agent should be deployed only as a decision-support recommendation for clinician review, given the ethical risks (patient safety, bias, explainability) discussed in the accompanying Solution document — it must never make autonomous treatment decisions.

6. Conclusion & Recommendations

- Deploy the post-pruned Decision Tree as an interpretable first-line diagnostic screening tool, given its strong recall and clinician-auditable rule structure.
- Use Logistic Regression's probability output for continuous risk stratification and prioritisation of follow-up testing.
- Pilot the Q-Learning treatment-recommendation agent in a clinician-in-the-loop advisory mode only, with mandatory human sign-off before any recommendation reaches a patient.
- Expand the training dataset with real, de-identified patient records and re-validate all models for demographic fairness before wider deployment.
- Establish a continuous monitoring pipeline to track model drift, False-Negative rates, and RL policy stability over time.